Differential Alternative Splicing in Plant Spaceflight Transcriptomes: A Multi-Study rMATS Analysis with Machine Learning and Ontology Integration

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Abstract

Spaceflight imposes unique mechanical and environmental stresses on plants, including microgravity, altered gravity regimes, ionizing radiation, and constrained growth conditions. While transcriptomic responses to spaceflight have been extensively characterized at the gene expression level, alternative splicing (AS)—a critical post-transcriptional regulatory mechanism—remains largely unexplored in the space biology context. We present a systematic analysis of differential alternative splicing across seven *Arabidopsis thaliana* RNA-seq studies from the NASA Open Science Data Repository (OSDR), using rMATS-turbo to quantify five AS event types (SE, A5SS, A3SS, MXE, RI) across diverse spaceflight-relevant stressors: partial and fractional gravity, ISS spaceflight, simulated galactic cosmic ray (GCR) radiation, and lunar regolith growth. Across all seven studies (199 samples, $\sim$100,000 total splicing events), retained intron (RI) events were the largest class of significant events in six of seven studies (51–72%; in OSD-37 only 10 events were significant, 3 of them RI). A frameshift heuristic that counts every RI event as NMD-sensitive classified 79–86% of significant events as potentially NMD-sensitive; this share largely reflects the RI fraction, and NMD itself was not measured. Circadian-related Gene Ontology terms were enriched (FDR $<$ 0.25, 1–4 genes each) in four of the five ISS flight datasets (OSD-314, OSD-120, OSD-37, OSD-678; OSD-120 and OSD-678 are root tips and leaves from the same CARA flight), while GCR radiation (OSD-658) enriched for phospholipid metabolism and stress-responsive translational regulation, lunar regolith (OSD-476, the most splicing-responsive study with 306 significant events in 257 genes) enriched for regulation of photosynthesis light reactions and of the mitotic cell cycle (FDR=0.138), and fractional gravity (OSD-251) enriched for superoxide response (FDR=0.001, FE=488$\times$). Cross-study recurrence analysis identified 93 genes significant in $\geq$2 studies, with AT4G17310 significant in four studies spanning gravity, radiation, and regolith stressors—the most recurrent target in this dataset, though not significant in the three ISS flight-versus-ground datasets. Seven additional genes (including SOS4/AT5G37850 and CRK28/AT4G21400) were significant in three studies each. Machine learning classification on the 199-sample, 9,875-gene PSI feature matrix achieved nested 5-fold CV AUC=0.598 for treatment-vs-control prediction, while 7-study leave-one-study-out cross-validation (LOSOCV) yielded near- or below-random performance (mean AUC 0.41–0.49 across Random Forest, Gradient Boosting and XGBoost). UMAP clustering confirmed that study-specific batch effects overwhelmingly dominate the splicing signal (ARI=0.995 for study,

−0.002 for treatment), a critical methodological finding for the field. **[TODO: no producing code or output in this repository]** A cross-species case study in *Brassica rapa* (OSD-59) identified 77 significant exon-skipping events. These findings establish alternative splicing as a previously underappreciated layer of plant spaceflight adaptation and are packaged with FAIR metadata for reuse by the space biology community (**[TODO: the analysis scripts are not yet in the repository]**).

Keywords: alternative splicing, spaceflight, microgravity, Arabidopsis, RNA-seq, rMATS, plant ontology, NASA OSDR, GeneLab

1 Introduction

Plants grown in spaceflight conditions must adapt to a fundamentally altered physical environment. Microgravity disrupts gravitropic signaling, modifies cell wall mechanics, and alters fluid dynamics, while the spaceflight environment additionally exposes organisms to ionizing radiation and constrained atmospheric conditions [1]. The transcriptomic responses of plants to spaceflight have been extensively documented through NASA's GeneLab program and the Open Science Data Repository (OSDR), with dozens of *Arabidopsis thaliana* spaceflight transcriptome studies now publicly available [2, 3] **[TODO: both references are unresolved; see reference list]**.

However, the vast majority of spaceflight transcriptomic analyses have focused on differential gene expression (DGE), treating each gene as a single transcriptional unit. This approach overlooks alternative splicing (AS), a post-transcriptional process that enables a single gene to produce multiple mRNA isoforms through differential exon inclusion, intron retention, and splice site selection [4]. In plants, AS is pervasive: over 60% of multi-exon *Arabidopsis* genes produce multiple transcript isoforms [5], and AS regulation is tightly coupled with developmental processes, hormone signaling, and stress responses [6, 7].

Alternative splicing is particularly relevant to spaceflight biology for several reasons. First, stress-induced intron retention—the most common AS response in plants under stress—frequently introduces premature termination codons (PTCs) that trigger nonsense-mediated decay (NMD), representing a rapid regulatory mechanism for downregulating gene expression without transcriptional changes [8, 9]. Second, AS can generate protein isoforms with altered functional domains, potentially modifying protein-protein interactions, enzymatic activity, or subcellular localization—all of which are relevant to mechanical stress adaptation [10]. Third, splicing factor genes themselves are subject to AS regulation, creating feedback loops that can amplify or modulate stress responses [11].

Despite this biological relevance, no systematic analysis of differential alternative splicing across multiple spaceflight studies has been published. The availability of pre-aligned STAR BAM files in the OSDR, processed through the GeneLab RNA-seq Consensus Pipeline (RCP, GL-DPPD-7101-E) using Ensembl Plants references [12], provides a unique opportunity to perform coordinated AS analysis across studies without the need for re-alignment.

Here we present a multi-study analysis of differential alternative splicing in plant spaceflight transcriptomes. We apply rMATS-turbo [13] to quantify five AS event types across proof-of-concept studies from the OSDR, spanning spaceflight, microgravity, Mars gravity, and cross-species comparisons. We integrate machine learning

classification, unsupervised clustering, and multi-ontology functional annotation (GO, Plant Ontology, Trait Ontology, MapMan) to characterize the functional landscape of spaceflight-induced splicing changes. Results are packaged with FAIR metadata for a planned Zenodo deposit (**[TODO: the analysis code is not yet in the repository]**).

2 Methods

2.1 Study Selection and Data Acquisition

Plant RNA-seq studies were enumerated from the NASA OSDR Biological Data API (BDAPI v2, `https://visualization.osdr.nasa.gov/biodata/api/v2`). We identified 58 plant RNA-seq studies, of which 54 were *Arabidopsis thaliana*. For this proof-of-concept analysis, we selected 8 studies (7 *Arabidopsis thaliana*, 1 *Brassica rapa*) representing diverse spaceflight-relevant conditions (Table 1).

All BAM files were aligned to Ensembl Plants release 48 references (TAIR10 for *Arabidopsis*, Brapa_1.0 for *Brassica*), matching the GeneLab RCP (GL-DPPD-7101-E) exactly, ensuring coordinate compatibility without re-alignment. OSDR records OSD-314 as a single-end library (50 bp) and the other six *Arabidopsis* studies as paired-end. For OSD-120, the coordinate-sorted BAMs used here were reported to lack mate-pair flags (flag=16, is_paired=False), so OSD-120 was also run in single-end mode (`-t single`); read type was auto-detected with pysam (**[TODO: the BAM flag inspection is not recorded in this repository]**).

Table 1: Proof-of-concept studies selected for analysis.

OSD ID	Organism	Tissue (OSDR)	Environment (OSDR)	BAMs
OSD-314	*A. thaliana*	Seedlings	ISS flight, μg and 0.34 g (centrifuge), vs 1 g on Earth	17
OSD-120	*A. thaliana*	Root tips	ISS spaceflight vs ground, CARA (3 genotypes $\times$ light/dark)	36
OSD-59	*B. rapa*	Leaf	ISS spaceflight vs ground	2
OSD-37	*A. thaliana*	Seedlings	ISS spaceflight vs ground (4 ecotypes)	56
OSD-678	*A. thaliana*	Leaves	ISS spaceflight vs ground, same CARA flight as OSD-120	36
OSD-658	*A. thaliana*	Whole seedlings	Simulated GCR radiation, 40 and 80 cGy (ground)	14
OSD-476	*A. thaliana*	Aboveground tissue	Apollo 11/12/17 regolith vs JSC-1A simulant (ground)	20
OSD-251	*A. thaliana*	Seedlings	ISS fractional gravity (μg–0.57 g vs in-flight 1 g), blue light	20

Tissue and design from the OSDR study records. **[TODO: OSD-59 outputs are not in this repository.]**

2.2 Differential Splicing Analysis

Differential alternative splicing was quantified using rMATS-turbo v4.3.0 [13], which detects and quantifies five AS event types: skipped exon (SE), alternative 5' splice site (A5SS), alternative 3' splice site (A3SS), mutually exclusive exons (MXE), and retained intron (RI). rMATS was run with junction count (JC) only mode, using the following parameters:

- Read type: single-end for OSD-314 (single-end library in OSDR) and OSD-120 (BAMs lacked mate-pair flags); paired-end for OSD-37, OSD-678, OSD-658, OSD-476 and OSD-251 (auto-detected from BAM flags via pysam)
- Library strandedness (OSDR): stranded for OSD-314, OSD-678, OSD-658 and OSD-476; unstranded for OSD-120, OSD-37 and OSD-251. **[TODO: the rMATS `--libType` used for each study is not recorded in this repository; an earlier draft stated fr-unstranded for all studies]**
- Read length (OSDR): 50 bp (OSD-314), 76 bp (OSD-120), 101 bp (OSD-37; one sample 151 bp), 151 bp (OSD-678, OSD-476), 75 bp (OSD-658), 125 bp (OSD-251). **[TODO: the `--readLength` passed to rMATS for each study is not recorded in this repository]**
- Variable read length: enabled (`--variable-read-length`)
- Statistical test: default rMATS-turbo likelihood ratio
- Threads: 8

Significance thresholds: FDR < 0.05 AND $|\Delta\text{PSI}| > 0.1$ (IncLevelDifference), as specified in the analysis plan.

2.3 NMD Prediction

Nonsense-mediated decay susceptibility was predicted for each significant event using a frameshift-based heuristic: for SE events, exon length not divisible by 3 predicts frameshift; for RI events, retained introns are classified as NMD-sensitive (intronic sequences typically contain stop codons); for A5SS/A3SS, length differences not divisible by 3 predict frameshift; for MXE, exon length differences not divisible by 3 predict frameshift. Because every RI event is classified as NMD-sensitive by rule, the NMD-sensitive share of significant events largely reflects the RI fraction; no transcript-level premature-termination-codon analysis or NMD-mutant data were used.

2.4 Machine Learning Analysis

Three complementary ML analyses were performed:

Flight-vs-ground classification (LOSOCV): Leave-one-study-out cross-validation using Random Forest, Gradient Boosting, and XGBoost classifiers. Feature selection (mutual information, top 500 features) was performed inside each fold to prevent leakage. SHAP values were computed for tree-based models to identify the most discriminative splicing events.

Nested k-fold cross-validation: Pooled 5-fold outer CV with 3-fold inner hyperparameter tuning (Random Forest), using all 199 samples from the seven *Arabidopsis* studies.

Unsupervised clustering: UMAP dimensionality reduction (n_neighbors=15 for multi-study) followed by K-means clustering, with evaluation by silhouette score and adjusted Rand index against known condition and study labels.

2.5 Ontology Enrichment Analysis

Over-representation analysis (ORA) was performed using Fisher's exact test with Benjamini-Hochberg FDR correction for Gene Ontology (MF/BP/CC, 251,290 annotations from TAIR/UniProt-GOA) and MapMan bins (24 bins, 11,793 genes mapped via GO-derived mapping). Plant Ontology and Trait Ontology enrichment modules were implemented but could not be run due to the lack of publicly available *Arabidopsis* gene-to-PO and gene-to-TO annotation files.

2.6 SpliceGrapher Visualization

SpliceGrapher v0.2.7 [14] was installed in an isolated Python 2.7 conda environment. Gene models were converted from Ensembl GTF to SpliceGrapher-compatible GFF format (filtered to gene/mRNA/exon records). Splice graphs were generated for the top 8 differentially spliced genes using `view_splicegraphs.py`.

3 Results

3.1 rMATS Detects Extensive Differential Splicing in Partial-Gravity Conditions

We applied rMATS-turbo to OSD-314, a study examining responses of *Arabidopsis* seedlings to altered gravity aboard the ISS (microgravity, and 0.34 g by centrifuge as a Mars-level gravity). Across 17 samples (11 flight samples: 5 microgravity and 6 at 0.34 g; 6 ground controls at 1 g on Earth), rMATS detected 10,532 total splicing events across five event types (Table 2). Of these, 161 events met our significance thresholds (FDR < 0.05, $|\Delta\text{PSI}| > 0.1$).

Table 2: rMATS event counts for OSD-314.

Event Type	Total	Significant	% Sig
SE (Skipped Exon)	2,602	12	0.46%
A5SS (Alt 5' SS)	1,319	9	0.68%
A3SS (Alt 3' SS)	1,960	32	1.63%
MXE (Mut. Excl. Exon)	50	0	0%
RI (Retained Intron)	4,601	108	2.35%
Total	**10,532**	**161**	**1.53%**

Retained intron events dominated the significant set (108/161 = 67%), consistent with the known role of intron retention as the primary AS response to stress in plants [8].

The ΔPSI range for significant events was [-0.659, 0.413], with 66 events showing increased inclusion in altered gravity and 95 showing decreased inclusion.

3.2 Top Differentially Spliced Genes Implicate Signaling and Metabolic Pathways

The most significant differentially spliced genes (Table 3) include several with known roles in stress signaling and metabolism. Notably, ECT5 (AT3G13060) is an m6A RNA reader involved in mRNA stability regulation, and its differential intron retention in altered gravity suggests post-transcriptional regulatory rewiring. CPK28 (AT5G66210) is a calcium-dependent protein kinase central to immune and stress signaling, and TPPA (AT5G51460) is annotated in Araport11 as a haloacid dehalogenase-like hydrolase (HAD) superfamily protein (UniProt O64896: trehalose-phosphate phosphatase A).

Table 3: Top differentially spliced events in OSD-314. Function: Araport11 description via Ensembl Plants.

Gene ID	Symbol	Event	FDR	ΔPSI	Function
AT3G13060	ECT5	RI	6.2e-11	-0.159	m6A RNA reader
AT1G07940	A1	SE	8.9e-11	-0.659	GTP-binding EF-Tu family
AT5G66210	CPK28	RI	2.8e-09	+0.142	Ca-dependent kinase
AT4G26600	—	RI	4.2e-08	-0.297	SAM-dependent methyltransferase
AT5G63160	BT1	RI	4.5e-08	+0.116	BTB/TAZ domain
AT5G51460	TPPA	A3SS	7.7e-08	-0.536	HAD-like hydrolase

3.3 Heuristic NMD Classification

Of the 161 significant events in OSD-314, 138 (86%) were classified as potentially NMD-sensitive by the frameshift heuristic. Because the heuristic counts all 108 RI events as NMD-sensitive, this proportion largely mirrors the RI share and does not show that the events trigger NMD. If confirmed by transcript-level premature-termination-codon analysis or NMD-mutant data (**[TODO: not done]**), it would be consistent with the "unproductive splicing" model proposed by Kalyna et al. [8], in which stress-induced intron retention serves as a post-transcriptional downregulation mechanism.

3.4 GO Enrichment Implicates Circadian, Light, and Stress Pathways

Gene Ontology over-representation analysis of the 144 unique genes with significant AS events (background: 60,985 annotated genes) returned 392 GO terms, of which 114 had $p < 0.05$ and 201 had FDR < 0.25. The top enriched terms include several with direct relevance to spaceflight conditions (Table 4).

The enrichment of circadian rhythm regulation (FDR=0.032) is particularly noteworthy, as spaceflight disrupts circadian signaling through altered light cycles and gravity perception. Shade avoidance and photoperiodism enrichment connects to the light

Table 4: Top enriched GO terms for genes with significant AS in OSD-314.

GO ID	Term	Genes	Fold Enr.	FDR
GO:0042754	Neg. regulation of circadian rhythm	2	141×	0.032
GO:0048574	Long-day photoperiodism, flowering	2	24×	0.649
GO:0009641	Shade avoidance	2	20×	0.613
GO:0032212	Telomere maintenance via telomerase	1	212×	0.205
GO:1901997	Neg. regulation of auxin biosynthesis	1	212×	0.205

regime differences (RED vs DARK) in the OSD-314 experimental design. Telomere maintenance enrichment may reflect radiation exposure effects, and auxin biosynthesis regulation connects to gravitropic signaling disruption.

3.5 Multi-Study Analysis: OSD-120 Extends Findings to ISS Spaceflight

To assess the generalizability of our findings, we analyzed OSD-120, a second *Arabidopsis* study examining root-tip responses to ISS spaceflight in the CARA experiment (18 flight vs 18 ground control samples, 3 genotypes $\times$ 2 light conditions; OSD-678 holds the leaves from the same flight). rMATS detected 13,570 total events, of which 32 were significant (Table 5). Retained intron events again dominated (23/32 = 72%), and 27/32 (84%) were predicted NMD-sensitive, closely mirroring the OSD-314 pattern.

Table 5: rMATS event counts for OSD-120.

Event Type	Total	Significant	% Sig
SE	4,761	4	0.08%
A5SS	1,561	1	0.06%
A3SS	2,249	4	0.18%
MXE	268	0	0%
RI	4,731	23	0.49%
Total	**13,570**	**32**	**0.24%**

The proportion of significant events was lower in OSD-120 than in OSD-314 (0.24% vs 1.53%). Both studies compare ISS flight samples with ground controls on Earth (OSDR lists the OSD-314 controls as '1G on Earth'), so the difference is not explained by a flight-versus-ground contrast. Tissue (root tips vs whole seedlings), library preparation and read length also differ (Table 1, Methods), and the cause was not tested here.

The OSD-120 GO table lists 114 terms, of which 55 had $p < 0.05$ and 82 FDR< 0.25. Notably, regulation of circadian rhythm was again enriched (FDR=0.097, fold enrichment=47×), along with histone H3K36/H3K9 demethylase activity (FDR=0.039), copper ion transport (FDR=0.031), and molybdenum cofactor metabolism (FDR=0.031). The convergence of circadian rhythm enrichment across both studies strengthens the connection between spaceflight and post-transcriptional regulation of clock genes.

3.6 OSD-37: High-Power Paired-End Analysis Reveals Small-Effect Splicing Changes

To extend our analysis to a larger, paired-end dataset, we analyzed OSD-37, an *Arabidopsis* study examining whole-seedling responses to ISS spaceflight across 4 ecotypes (Ws-2, Ler-0, Cvi-0, Col-0; 56 paired-end BAMs: 28 flight vs 28 ground). rMATS detected 20,590 total events—the largest event count of the seven *Arabidopsis* studies—with a more balanced event-type distribution than OSD-314 or OSD-120 (Table 6).

Table 6: rMATS event counts for OSD-37.

Event Type	Total Events	FDR<0.05	FDR<0.05 & $\|\Delta\text{PSI}\| > 0.1$
SE	10,545	34	4
A5SS	1,832	23	1
A3SS	2,517	19	0
MXE	685	25	2
RI	5,011	74	3
Total	**20,590**	**175**	**10**

The striking discrepancy between the 175 events reaching FDR < 0.05 and the 10 meeting the combined $|\Delta\text{PSI}| > 0.1$ threshold reflects the high statistical power afforded by 56 samples. The median $|\Delta\text{PSI}|$ among FDR-significant events was only 0.025 (range: 0.001–0.234), indicating that the vast majority of statistically significant splicing changes in this large cohort are biologically small. This highlights the importance of effect-size thresholds in large-sample splicing studies.

The 10 biologically significant events (8 NMD-sensitive, 80%) included several genes with direct relevance to spaceflight adaptation: **RVE1** (AT5G17300, REVEILLE 1, a circadian clock MYB-like transcription factor with both SE and A5SS events, ΔPSI=-0.111 and -0.110), **SR45A** (AT1G07350, a serine/arginine-rich splicing regulator, SE event ΔPSI=-0.153), **CAS** (AT5G23060, calcium-sensing protein, RI event ΔPSI=-0.183), **HHP2** (AT4G30850, heptahelical transmembrane protein 2, SE event ΔPSI=-0.234), and **CRK28** (AT4G21400, cysteine-rich receptor-like kinase, RI event ΔPSI=0.110). GO enrichment (46 terms returned; 20 at $p < 0.05$, 29 at FDR< 0.25) implicated auxin-activated signaling pathway (FDR=0.065, FE=35$\times$), cellular response to calcium ion (FDR=0.020, FE=452$\times$), regulation of RNA splicing (FDR=0.028, FE=205$\times$), and regulation of stomatal closure (FDR=0.063, FE=72$\times$).

3.7 OSD-678: ISS Spaceflight with Photoreceptor Mutants Reveals Strong Splicing Response

To further extend the cross-study analysis, we analyzed OSD-678, the leaf transcriptomes from the same CARA flight as OSD-120 (3 genotypes: Col-0, Ws, phyD; 2 light conditions: dark, light; 36 paired-end BAMs, 18 flight vs 18 ground). rMATS detected 15,255 total events with 190 significant under the combined threshold (FDR < 0.05, $|\Delta\text{PSI}| > 0.1$)—the second-highest significant event count after OSD-476 (Table 7).

As in five of the other six *Arabidopsis* studies, retained intron events were the largest class (96/190, 51%), and the NMD heuristic classified 154/190 (81%) as po-

Table 7: rMATS event counts for OSD-678.

Event Type	Total Events	FDR<0.05	FDR<0.05 & $\|\Delta\text{PSI}\| > 0.1$
SE	5,918	264	22
A5SS	1,704	168	25
A3SS	2,423	255	43
MXE	266	25	4
RI	4,944	793	96
Total	**15,255**	**1,505**	**190**

tentially NMD-sensitive (79–86% across the seven studies). The 190 significant events mapped to 156 unique genes (84 up, 106 down). GO enrichment (356 terms returned; 101 at $p < 0.05$, 171 at FDR< 0.25) again implicated **regulation of circadian rhythm** (FDR=0.008, FE=18.6×, 4 genes)—the fourth dataset with a circadian-related term at FDR $<$ 0.25, although OSD-678 and OSD-120 come from the same flight—alongside nuclear speck (FDR=0.024, FE=8.5×, reflecting splicing machinery self-regulation), chloroplast localization, CAAX-box protein maturation, and negative regulation of innate immune response. Circadian-related terms thus recur across four flight datasets (three independent flight experiments), each resting on 1–4 genes.

3.8 OSD-658: Simulated Galactic Cosmic Ray Radiation Activates Stress Splicing

To extend our analysis beyond gravity-related stressors, we analyzed OSD-658, a ground-based *Arabidopsis* study of whole seedlings exposed to simulated galactic cosmic ray (GCR) radiation (14 paired-end BAMs: 8 irradiated [4 at 40 cGy GCR + 4 at 80 cGy GCR] vs 6 non-irradiated controls). rMATS detected 9,932 total events with 43 significant under the combined threshold (Table 8).

Table 8: rMATS event counts for OSD-658.

Event Type	**Total Events**	**Significant (FDR<0.05 & $\|\Delta\text{PSI}\| > 0.1$)**
SE	2,125	2
A5SS	1,302	7
A3SS	1,915	9
MXE	45	0
RI	4,545	25
Total	**9,932**	**43**

Retained intron events again dominated (25/43 = 58%), and 34/43 (79%) were predicted NMD-sensitive—the lowest NMD proportion across the seven studies but still substantial. The 43 events mapped to 38 unique genes (30 up, 13 down). GO enrichment (156 terms returned; 81 at $p < 0.05$, 115 at FDR< 0.25) implicated response to herbicide (FDR=0.009, FE=180×), phospholipid metabolic process (FDR=0.013, FE=104×), and stress-responsive translational regulation (FDR=0.016, FE=989×), consistent with the oxidative and genotoxic stress imposed by ionizing radiation.

The top differentially spliced genes included AT5G13790 (A3SS, ΔPSI=$-$0.485), AT4G25990 (RI, ΔPSI=+0.476), and AT4G17310 (RI, ΔPSI=+0.416)—notably, AT4G17310 emerged as the most recurrent differential splicing target across all seven studies (see below).

3.9 OSD-476: Lunar Regolith Elicits the Largest Splicing Response

To examine splicing responses to an extraterrestrial substrate, we analyzed OSD-476, an *Arabidopsis* study profiling aboveground tissue of plants grown in Apollo lunar regolith (Apollo 11, 12, and 17 samples; 12 BAMs) versus JSC-1A lunar simulant control (8 BAMs). rMATS detected 14,766 total events with 306 significant—the largest significant event count of any study in our analysis (Table 9).

Table 9: rMATS event counts for OSD-476.

Event Type	**Total Events**	**Significant (FDR$<$0.05 & $\lvert\Delta$PSI$\rvert > 0.1$)**
SE	5,428	33
A5SS	1,678	30
A3SS	2,467	64
MXE	259	2
RI	4,934	177
Total	**14,766**	**306**

Retained intron events dominated (177/306 = 58%), and 263/306 (86%) were NMD-sensitive. The 306 events mapped to 257 unique genes (199 up, 107 down)—the largest gene set of any study. GO enrichment (485 terms returned; 114 at $p < 0.05$, 181 at FDR< 0.25) implicated blue light photoreceptor activity (FDR=0.138, FE=33$\times$), regulation of photosynthesis light reactions (FDR=0.138, FE=29$\times$), regulation of mitotic cell cycle (FDR=0.138, FE=23$\times$), and telomeric chromosome region (FDR=0.138, FE=21$\times$). The photosynthesis-related enrichment is consistent with the known nutrient and physical limitations of lunar regolith as a plant growth substrate. The top differentially spliced genes included AT2G47020 (RI, ΔPSI=+0.475), AT1G06710 (A5SS, ΔPSI=+0.377), and AT2G30370 (RI, ΔPSI=$-$0.361).

3.10 OSD-251: Fractional Gravity Under Blue Light

To further probe gravity-dose-dependent splicing, we analyzed OSD-251, an *Arabidopsis* spaceflight study of seedlings grown at fractional gravity levels on the ISS under blue light (all samples in flight; 20 paired-end BAMs: 17 altered gravity [uG, 0.09G, 0.18G, 0.36G, 0.57G] vs 3 in-flight 1G controls). rMATS detected 15,220 total events with 86 significant (Table 10).

Retained intron events dominated (49/86 = 57%), and 73/86 (85%) were NMD-sensitive. The 86 events mapped to 77 unique genes (51 up, 35 down). GO enrichment (239 terms returned; 93 at $p < 0.05$, 170 at FDR< 0.25) implicated response to superoxide (FDR=0.001, FE=488$\times$)—the strongest single GO term across all seven studies—alongside nuclear speck (FDR=0.061, FE=12$\times$) and positive regulation of

Table 10: rMATS event counts for OSD-251.

Event Type	Total Events	Significant (FDR$<$0.05 & $\lvert\Delta$PSI$\rvert > 0.1$)
SE	5,940	9
A5SS	1,714	15
A3SS	2,392	13
MXE	323	0
RI	4,851	49
Total	**15,220**	**86**

mRNA splicing via spliceosome (FDR=0.079, FE=244$\times$). The superoxide response enrichment connects splicing changes to oxidative stress, a known consequence of altered gravity conditions. The top differentially spliced genes included AT2G16940 (SE, ΔPSI=+0.417), AT3G09370 (RI, ΔPSI=+0.395), and AT1G10240 (RI, ΔPSI=$-$0.347).

3.11 Cross-Study Recurrence Identifies Shared Splicing Targets Across Diverse Stressors

Comparison of significant genes across all seven *Arabidopsis* studies (OSD-314: 144 genes, OSD-120: 31, OSD-37: 9, OSD-678: 156, OSD-658: 38, OSD-476: 257, OSD-251: 77) revealed 93 recurrent genes significant in $\geq$2 studies (Table 11). The most striking finding is **AT4G17310**, significant in four studies (OSD-251, OSD-314, OSD-476, OSD-658)—spanning fractional gravity, partial gravity, lunar regolith, and GCR radiation. Seven additional genes were significant in three studies each.

Table 11: Cross-study recurrence of significant genes (7 *Arabidopsis* studies).

Gene	n_studies	Studies	Ensembl description
AT4G17310	4	OSD-251, OSD-314, OSD-476, OSD-658	Uncharacterized protein
AT5G57565	3	OSD-251, OSD-314, OSD-678	Protein kinase superfamily
AT5G37850 (SOS4)	3	OSD-314, OSD-476, OSD-678	pfkB-like kinase (pyridoxal kinase)
AT4G16660	3	OSD-314, OSD-476, OSD-678	Hsp70 family protein
AT2G20815 (QWRF3)	3	OSD-251, OSD-314, OSD-678	QWRF motif protein (DUF566)
AT4G21400 (CRK28)	3	OSD-37, OSD-476, OSD-678	Cysteine-rich RLK 28
AT5G64460	3	OSD-251, OSD-476, OSD-678	Phosphoglycerate mutase family
AT5G18245	3	OSD-251, OSD-476, OSD-678	ncRNA

The largest pairwise overlaps were OSD-678 $\cap$ OSD-476 (26 genes) and OSD-314 $\cap$ OSD-476 (22 genes), both exceeding the previously identified OSD-314 $\cap$ OSD-

678 overlap (18 genes). **[TODO: 4-study analysis: no producing code or output in this repository]** The expansion from 4 to 7 studies increased the recurrent gene set from 31 to 93 genes, and SOS4 (AT5G37850, pyridoxal kinase) and CRK28 (AT4G21400, cysteine-rich receptor-like kinase) expanded their recurrence from 2 to 3 studies. AT4G17310 is the most recurrent gene in the 7-study set, but its event type and direction differ between studies (RI in OSD-314 and OSD-658, SE in OSD-251 and OSD-476; $\triangle$PSI -0.171, $+0.416$, -0.155 and -0.132), and it is not significant in the three ISS flight-versus-ground datasets (OSD-120, OSD-37, OSD-678). The 7-study feature matrix spanned 9,875 unique genes across 199 samples.

3.12 Machine Learning Classification: 7-Study Analysis Confirms Batch Effect Dominance

We constructed a unified feature matrix of 199 samples (56 from OSD-37, 36 from OSD-120, 36 from OSD-678, 20 from OSD-476, 20 from OSD-251, 17 from OSD-314, 14 from OSD-658) $\times$ 9,875 genes (gene-level mean PSI, filtered for $\geq$10 samples, mean-imputed; 112 treatment, 87 control). The binary classification task (treatment vs control, encompassing flight, radiation, regolith, and altered gravity as treatment) was evaluated using two strategies:

Leave-one-study-out cross-validation (LOSOCV, 7-study): With 7 studies, each fold trains on 6 studies and tests on the held-out study. All models performed near or below random (mean $\pm$ SD across the seven held-out studies): RF AUC=0.493 $\pm$ 0.127, GBM AUC=0.423 $\pm$ 0.116, XGBoost AUC=0.412 $\pm$ 0.112. The best individual-fold performance was RF on OSD-314 (AUC=0.667) and RF on OSD-476 (AUC=0.635). **[TODO: 4-study analysis: no producing code or output in this repository]** The addition of three studies with distinct stressor types (radiation, regolith, fractional gravity) further reduced cross-study generalization compared to the 4-study analysis, as the heterogeneity of stress conditions increased the diversity of study-specific batch signatures.

Nested 5-fold cross-validation (pooled, 7 studies): When samples from all seven studies were pooled with 5-fold CV (inner hyperparameter tuning), Random Forest achieved AUC=0.598 $\pm$ 0.084 (per-fold: 0.707, 0.624, 0.578, 0.606, 0.476). This represents a decrease from the 4-study pooled AUC of 0.698 $\pm$ 0.046 (**[TODO: no producing code or output in this repository]**), reflecting the increased heterogeneity of stressor types (radiation, regolith, and fractional gravity introduce splicing patterns that differ from ISS flight responses). The within-pooled discriminative signal remains above random but is less stable, with one fold dropping to AUC=0.476.

SHAP feature importance analysis identified the top splicing-based predictors: AT1G56220 (mean $|$SHAP$|$=0.0027), AT5G01770 (0.0019), AT1G48360 (0.0018), AT3G15620 (0.0016), and AT3G62190 (0.0016). The SHAP ranking changed substantially from the 4-study analysis (**[TODO: no producing code or output in this repository]**), reflecting the dominance of study-specific batch effects: different genes become discriminative depending on study composition, and no single gene maintains consistent SHAP importance across analysis iterations.

3.13 Multi-Study UMAP Clustering Reveals Study-Specific Batch Effects

UMAP dimensionality reduction of the 199-sample, 7-study feature matrix (n_neighbors=15) revealed that samples clustered almost perfectly by study identity (ARI=0.995) with a silhouette score of 0.926, while treatment/control status showed no clustering signal (ARI=$-$0.002). The near-perfect ARI for study identity (up from 0.864 in the 4-study analysis; **[TODO: no producing code or output in this repository]**) indicates that adding three more studies with distinct stressor types further separated the study clusters, as each stressor type produces a more distinct splicing signature. This is a critical finding for spaceflight splicing research: study-specific batch effects—arising from different hardware, growth chambers, sequencing protocols, accession/genotype backgrounds, stressor types, and experimental timelines—overwhelmingly dominate the splicing signal and obscure the treatment/control distinction. This explains the near-random LOSOCV performance and underscores the need for cross-study batch correction (e.g., ComBat) or study-stratified analysis in future multi-study splicing investigations.

3.14 Cross-Study GO Enrichment Comparison Reveals a Core Splicing-Associated Signature

To systematically compare the stressor-specific functional profiles across all seven studies, we constructed a unified GO enrichment matrix from the 748 unique GO terms significant at FDR $<$ 0.25 across studies (OSD-314: 201, OSD-120: 82, OSD-37: 29, OSD-678: 171, OSD-658: 115, OSD-476: 181, OSD-251: 170). Four complementary visualizations were generated (Figures 8–11; **[TODO: no producing code or output in this repository; these figures are not included]**).

A GO term $\times$ study heatmap (Figure 8) of the top 50 terms revealed both shared and stressor-specific enrichment patterns. The upset plot (Figure 9) quantified the overlap structure: 157 GO terms were significant in $\geq$2 studies, 31 in $\geq$3 studies, and 9 in $\geq$4 studies. The single most recurrent term was **nuclear speck (GO:0016607)**, significant in 6 of 7 studies (all except OSD-658), consistent with involvement of spliceosomal machinery in the splicing response itself. Three terms were significant in $\geq$5 studies: nuclear speck, nucleic acid binding (GO:0003676, 5 studies), and protein binding (GO:0005515, 5 studies).

The 9 core terms significant in $\geq$4 studies (Table 12) were subjected to GO semantic similarity analysis using the Wang method with Arabidopsis annotation (Figure 10; **[TODO: no producing code or output in this repository]**). The dendrogram revealed three functional clusters: (1) **RNA processing**—nuclear speck (GO:0016607), nucleus (GO:0005634), and RNA splicing (GO:0008380), reflecting the spliceosomal core; (2) **binding/catalytic**—nucleotide binding (GO:0000166), nucleic acid binding (GO:0003676), and protein binding (GO:0005515); and (3) **stress physiology**—circadian rhythm (GO:0007623), Mo-molybdopterin cofactor biosynthesis (GO:0006777), and abscisic acid biosynthesis (GO:0009688). The clustering of circadian rhythm with hormone biosynthesis rather than with RNA processing suggests that the shared splicing response targets regulatory physiology rather than merely the splicing machinery itself.

A stressor-grouped dot plot (Figure 11; **[TODO: no producing code or output**

Table 12: Core GO terms significant (FDR < 0.25) in $\geq$4 of 7 studies, generated from the seven `OSD-*_go_enrichment.tsv` files.

GO ID	Term	Ontology	n_studies	Studies
GO:0016607	nuclear speck	CC	6	314, 120, 37, 678, 476, 251
GO:0003676	nucleic acid binding	MF	5	314, 37, 678, 476, 251
GO:0005515	protein binding	MF	5	314, 120, 37, 678, 251
GO:0000166	nucleotide binding	MF	4	120, 678, 658, 251
GO:0005634	nucleus	CC	4	314, 658, 476, 251
GO:0006777	Mo-molybdopterin cofactor biosynthetic process	BP	4	120, 678, 658, 476
GO:0007623	circadian rhythm	BP	4	314, 120, 37, 476
GO:0008380	RNA splicing	BP	4	37, 678, 476, 251
GO:0009688	abscisic acid biosynthetic process	BP	4	314, 120, 658, 476

in this repository]) of the top 5 terms per study (by fold enrichment) highlighted the stressor-specific signatures: ISS flight studies enriched for circadian rhythm and photoperiodism; OSD-658 (radiation) for phospholipid metabolism and translational stress; OSD-476 (regolith) for blue light photoreceptor activity and photosynthesis regulation; and OSD-251 (fractional gravity) for superoxide response (FDR=0.001, FE=488$\times$, the strongest single GO term across all studies). The largest pairwise GO overlaps were OSD-314 $\cap$ OSD-476 (32 terms), OSD-314 $\cap$ OSD-678 (28 terms), and OSD-476 $\cap$ OSD-678 (25 terms), consistent with the gene-level recurrence pattern.

3.15 Batch Correction Improves Cross-Study ML Generalization

[TODO: no producing code or output in this repository for any result in this subsection (Table 13, Figures 12–14). The uncorrected-matrix row also disagrees with the 7-study results above (LOSOCV RF/GBM/XGBoost 0.493/0.423/0.412; nested k-fold 0.598 $\pm$ 0.084).]

To directly address the study-specific batch effects identified in the UMAP analysis, we applied two batch correction methods to the 199 $\times$ 9,875 PSI feature matrix: ComBat (parametric empirical Bayes, via sva) and limma::removeBatchEffect (non-parametric linear model). Both methods used study identity as the batch variable and preserved the treatment (is_flight) covariate via a design matrix, removing study-specific variation while retaining the treatment signal. Corrected values were clipped to the [0, 1] PSI range. ComBat identified 5,580 genes with uniform expression within a single batch (all zeros) that were not adjusted.

The three matrices (uncorrected, ComBat, limma) were evaluated using the full ML pipeline (LOSOCV, nested 5-fold CV, UMAP, SHAP). The UMAP analysis provided the most reliable assessment of batch structure removal (Table 13, Figure 12): limma::removeBatchEffect nearly eliminated study-specific clustering (ARI(study) = 0.021, down from 0.995), while ComBat substantially reduced it (ARI(study) = 0.524). Both methods marginally increased the treatment clustering signal (ARI(treatment): uncorrected = -0.002, ComBat = 0.003, limma = 0.004), though the treatment signal remained negligible.

LOSOCV performance improved with both correction methods (Figure 13): RF

Table 13: Batch correction ML comparison (3 matrices). **[TODO: no producing code or output in this repository]**

Matrix	LOSOCV RF	LOSOCV GBM	LOSOCV XGB	Nested k-fold	ARI(study)	ARI(treat)
Uncorrected	0.475±0.082	0.507±0.150	0.491±0.172	0.613±0.037	0.995	−0.002
ComBat	0.798±0.256	0.588±0.317	0.705±0.328	1.000*	0.524	0.003
limma	0.553±0.056	0.581±0.152	0.614±0.127	0.656±0.053	0.021	0.004

*ComBat nested k-fold AUC = 1.0 is inflated by data leakage: batch correction was applied to the full matrix before CV splitting. LOSOCV and UMAP ARI are more reliable.

mean AUC increased from 0.475 (uncorrected) to 0.798 (ComBat) and 0.553 (limma). However, ComBat's improvement showed high variance (±0.256), with AUC=1.0 for OSD-251, OSD-476, and OSD-678 but AUC=0.39–0.49 for OSD-314 and OSD-37, indicating that ComBat over-corrects for some studies while under-correcting for others. limma produced more stable but modest improvements (±0.056). The nested k-fold AUC for limma (0.656 ± 0.053) represents a modest but genuine improvement over the uncorrected baseline (0.613 ± 0.037), while the ComBat nested k-fold AUC of 1.0 reflects data leakage from applying batch correction before cross-validation splitting.

SHAP feature importance analysis (Figure 14) revealed that AT1G56220—the top predictor in the uncorrected matrix—remained the second-ranked predictor in the limma-corrected matrix, suggesting it captures a genuine treatment-associated splicing signal rather than a study-specific artifact. The ComBat-corrected SHAP ranking was entirely different (top: AT5G07090, AT5G39740, AT1G56280), consistent with the more aggressive batch structure removal altering the discriminative feature landscape.

3.16 Cross-Species Comparison: *Brassica rapa*

[TODO: no producing code or output in this repository] Analysis of OSD-59 (*Brassica rapa*, ISS spaceflight) with only 2 BAMs (1 flight, 1 ground) detected 1,833 SE events (71 significant) and 44 MXE events (6 significant), for a total of 77 significant events. The limited sample size precluded detection of A5SS, A3SS, and RI events (rMATS requires replicates for variance estimation). Despite this limitation, the detection of significant exon-skipping events in *Brassica* demonstrates that spaceflight-induced alternative splicing is conserved across species, supporting the broader applicability of our pipeline.

4 Discussion

Our analysis reveals that alternative splicing is a substantial and previously underappreciated component of the plant spaceflight transcriptomic response. The predominance of retained intron events (the largest class in six of seven studies, 51–72% of significant events) is consistent with the broader plant stress biology literature, where intron retention serves as a regulatory mechanism for transcriptome reprogramming [8, 9]. The 79–86% NMD-sensitive share comes from a heuristic that counts every RI event as NMD-sensitive, so it largely restates the RI fraction; whether these transcripts are NMD targets was not tested. The pattern recurs across seven datasets from six independent experiments (OSD-120 and OSD-678 are root tips and leaves

from the same CARA flight), with different designs, sample sizes, sequencing protocols and stressors (gravity, radiation, regolith). This is consistent with intron retention being the predominant AS change under these conditions.

The enrichment of circadian rhythm, photoperiodism, and shade avoidance pathways among genes with significant AS events is particularly relevant to spaceflight conditions. The ISS environment imposes artificial light cycles that differ from natural photoperiods, and microgravity disrupts the gravitropic signaling that normally coordinates growth with light perception. Our finding that splicing of circadian regulators is altered in partial-gravity conditions (OSD-314, FDR=0.032), ISS spaceflight (OSD-120, FDR=0.097), again in OSD-37 (where RVE1, a core circadian clock transcription factor, showed significant SE and A5SS events), and most strongly in OSD-678 (FDR=0.008, FE=18.6$\times$, where LHY—the other core circadian clock MYB transcription factor—was itself a recurrent differential splicing target with OSD-314) suggests a post-transcriptional layer to the known spaceflight-induced circadian disruption [15]. Circadian-related terms were enriched in four of the five flight datasets (not in OSD-251). Because OSD-120 and OSD-678 come from the same CARA flight, this is three independent flight experiments, and each enrichment rests on 1–4 genes; it is a recurring pattern to test rather than an established finding. The two ground-based studies (OSD-658 radiation, OSD-476 regolith) and the fractional-gravity flight study OSD-251 showed distinct GO enrichment profiles—phospholipid metabolism and translational stress for radiation, photosynthesis and cell cycle for regolith (FDR=0.138), and superoxide response for fractional gravity—while sharing the RI-dominant splicing pattern.

The identification of CPK28 (calcium-dependent protein kinase) as a differentially spliced gene in OSD-314, and CAS (calcium-sensing protein) and CRK28 (cysteine-rich receptor-like kinase) in OSD-37, connects AS to calcium signaling, which is itself disrupted by microgravity. Calcium signaling is central to gravitropic response, and post-transcriptional regulation of these signaling components via intron retention may represent a feedback mechanism modulating gravitropic sensitivity under altered gravity conditions. The expansion of CRK28 (AT4G21400) recurrence to three studies (OSD-37, OSD-476, OSD-678) in the 7-study analysis further reinforces this connection, suggesting that receptor-like kinase splicing regulation is a shared response across gravity, regolith, and flight stressors.

The OSD-37 analysis provides an important methodological insight: with 56 samples, rMATS detected 175 events at FDR < 0.05, but only 10 (5.7%) met the biologically meaningful $|\Delta\text{PSI}| > 0.1$ threshold. The median $|\Delta\text{PSI}|$ among FDR-significant events was 0.025, indicating that large-sample splicing studies detect predominantly small-effect changes. This has implications for the design of spaceflight splicing studies: statistical significance alone is insufficient, and effect-size thresholds are essential to distinguish biologically meaningful splicing changes from the background of subtle, high-power detections. This also explains the lack of cross-study gene overlap with OSD-37—the stringent combined threshold identifies only the largest-effect events, which are study-specific.

The 7-study cross-study recurrence analysis revealed 93 genes significant in $\geq$2 studies, a substantial expansion from the 31 recurrent genes identified in the 4-study analysis (**[TODO: no producing code or output in this repository]**). The most striking finding is AT4G17310, significant in four studies (OSD-251, OSD-314, OSD-476, OSD-658) spanning fractional gravity, partial gravity, lunar regolith, and GCR radiation—the most recurrent target in this dataset, although its event type and di-

rection differ between studies and it is not significant in the three ISS flight-versus-ground datasets. Seven additional genes were significant in three studies each, including SOS4 (AT5G37850, pyridoxal kinase involved in stress responses), CRK28 (AT4G21400, receptor-like kinase), and QWRF3 (AT2G20815). The largest pairwise overlaps were OSD-678 ∩ OSD-476 (26 genes) and OSD-314 ∩ OSD-476 (22 genes), both exceeding the OSD-314 ∩ OSD-678 overlap (18 genes). The expansion of recurrence with additional studies demonstrates that while individual significant events are largely study-specific, a core set of stress-responsive splicing targets emerges when diverse stressors are compared. Notably, several recurrent genes showed opposite ΔPSI directions between studies, suggesting context-dependent splicing regulation shaped by the distinct experimental conditions.

The machine learning results provide important insights into the discriminative power and limitations of splicing profiles. The 7-study nested 5-fold CV AUC of 0.598 ± 0.084 demonstrates that splicing variation carries moderate signal for treatment-vs-control classification when evaluated within a pooled dataset, but this signal decreased from the 4-study AUC of 0.698 ± 0.046, reflecting the increased heterogeneity of stressor types (**[TODO: the 4-study values 0.698 and 0.864 in this paragraph have no producing code or output in this repository]**). The 7-study LOSOCV performance collapsed to near- or below-random (mean AUC 0.41–0.49 across models), confirming that cross-study generalization fails when studies encompass fundamentally different stressors. Critically, the UMAP analysis revealed that study identity almost perfectly separated samples (ARI=0.995 in the 7-study analysis, up from 0.864 in the 4-study analysis) while treatment/control status did not (ARI=-0.002), confirming that study-specific batch effects overwhelmingly dominate the splicing signal. The increase in ARI(study) from 0.864 to 0.995 upon adding three studies with distinct stressor types indicates that each stressor type produces a more distinct splicing signature, further separating study clusters. This is a key finding for the field: cross-study splicing comparisons require explicit batch correction or study-stratified analysis, and the treatment/control splicing signal is embedded within a larger study-specific signature.

The SHAP-identified top predictors in the 7-study analysis (AT1G56220, AT5G01770, AT1G48360, AT3G15620, AT3G62190) differ substantially from both the 3-study and 4-study rankings (**[TODO: no producing code or output in this repository]**), reflecting the dominance of study-specific batch effects: different genes become discriminative depending on study composition, and no single gene maintains consistent SHAP importance across analysis iterations. This instability is itself informative—it confirms that the ML models are learning study-specific rather than stressor-generic splicing patterns. **[TODO: batch-correction results: no producing code or output in this repository]** However, the batch correction analysis provides a critical nuance: AT1G56220 remained the second-ranked SHAP predictor in the limma-corrected matrix, suggesting that at least one gene captures a genuine treatment-associated signal that survives batch structure removal. Further functional characterization of the recurrent differential splicing targets (particularly AT4G17310, SOS4, and CRK28) in the context of spaceflight is warranted.

[TODO: batch-correction results: no producing code or output in this repository] The batch correction analysis directly addressed the study-specific batch effects identified in the UMAP analysis. Two methods—ComBat (parametric empirical Bayes) and limma::removeBatchEffect (non-parametric linear model)—

were applied with treatment preserved as a covariate and study as the batch variable. limma::removeBatchEffect nearly eliminated study-specific clustering (UMAP ARI(study) = 0.021, down from 0.995), while ComBat substantially reduced it (ARI = 0.524). Both methods produced modest improvements in LOSOCV classification (RF AUC: uncorrected 0.475 $\rightarrow$ ComBat 0.798 $\rightarrow$ limma 0.553), though ComBat's improvement was unstable across held-out studies ($\pm$0.256). The nested k-fold AUC for limma (0.656 $\pm$ 0.053) represents a genuine but modest improvement over the uncorrected baseline (0.613 $\pm$ 0.037). Critically, the ComBat nested k-fold AUC of 1.0 is inflated by data leakage—batch correction was applied to the full matrix before cross-validation splitting—and should not be interpreted as true generalization performance. This highlights a methodological consideration for future batch-corrected ML analyses: batch correction must be applied within each CV fold (fit on training data only, transform test data) to avoid optimistic bias. The persistence of AT1G56220 as a top SHAP predictor in the limma-corrected matrix, despite the near-complete removal of study structure, suggests it may represent a genuine treatment-associated splicing signal worthy of experimental validation. The cross-study GO comparison further revealed that the shared splicing response converges on nuclear speck (GO:0016607, 6/7 studies), RNA splicing (GO:0008380, 4/7 studies), and circadian rhythm (GO:0007623, 4/7 studies), establishing a core post-transcriptional regulatory signature that transcends individual stressor types.

5 Limitations

This proof-of-concept analysis has several limitations. First, OSD-314 is a single-end library and the OSD-120 BAMs lacked mate-pair flags, so both were run in single-end mode, while the other five studies were run paired-end. While this does not affect junction-based event detection, single-end mode may reduce sensitivity for events that benefit from paired-end read evidence in OSD-314 and OSD-120. **[TODO: the rMATS library-type (`--libType`) and read-length settings used for each study are not recorded in this repository; OSD-314, OSD-678, OSD-658 and OSD-476 are stranded libraries per OSDR, and an earlier draft stated fr-unstranded for all studies.]** Second, the OSD-59 *Brassica* analysis was limited to 1 sample per condition, precluding variance estimation for most event types. Third, the MapMan bin mapping was derived from GO annotations rather than the official MapMan mapping files (which require a license agreement), potentially reducing bin assignment accuracy. Fourth, Plant Ontology and Trait Ontology enrichment could not be performed due to the lack of publicly available *Arabidopsis* gene-to-PO and gene-to-TO annotation files. Fifth, the 7-study LOSOCV classification yielded near- or below-random performance (mean AUC 0.41–0.49 across models), and the UMAP analysis confirmed that study-specific batch effects dominate the splicing signal (ARI(study)=0.995). **[TODO: batch-correction results: no producing code or output in this repository]** We applied two batch correction methods (ComBat and limma::removeBatchEffect) with treatment preserved as a covariate; limma nearly eliminated study-specific structure (ARI(study)=0.021) and produced modest LOSOCV improvement (RF AUC 0.553), while ComBat showed higher but unstable improvement (RF AUC 0.798 $\pm$ 0.256). However, the ComBat nested k-fold AUC of 1.0 is inflated by data leakage (batch correction applied before CV splitting), and a properly nested batch correction (fit within each CV fold) would be

needed for unbiased generalization estimates. Expansion to 8+ studies with fold-aware batch correction would further improve cross-study generalization assessment. Sixth, the cross-study gene overlap is modest (93 recurrent genes across 7 studies, with the largest pairwise overlap of 26 genes between OSD-678 and OSD-476), reflecting the heterogeneity of spaceflight experimental designs, the stringent significance thresholds, and the different effect-size regimes across studies. Seventh, the OSD-37 analysis revealed that with 56 samples, 175 events reach FDR < 0.05 but only 10 meet $|\Delta\text{PSI}| > 0.1$, highlighting the challenge of distinguishing biologically meaningful from statistically significant splicing changes in large cohorts. Finally, the SHAP-identified biomarker genes require experimental validation in independent datasets.

6 Data Availability

All RNA-seq data analyzed in this study is publicly available from the NASA Open Science Data Repository (OSDR) at `https://osdr.nasa.gov`. Results and the FAIR deposit metadata are available at `https://github.com/dr-richard-barker/OSDR_Plant_Alternative_Splicing`. **[TODO: Zenodo DOI; the analysis pipeline code is not yet in the repository.]**

7 Code Availability

[TODO: The analysis scripts are not yet in the repository at `https://github.com/dr-richard-barker/OSDR_Plant_Alternative_Splicing`; they must be added before submission.] The pipeline uses rMATS-turbo v4.3.0 (conda installable from bioconda), Python 3.10+, SpliceGrapher 0.2.7 (Python 2.7), and standard scientific Python packages.

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References

[1] Paul AL, et al. (2012) Spaceflight transcriptomes: unique responses to a novel environment. *Astrobiology* 12(1): 40–56. doi:10.1089/ast.2011.0696

[2] **[TODO: unresolved citation. The original entry (Beheshti A, et al. (2018) NASA GeneLab: interfaces for the exploration of spaceflight and ground-based omics data. Front Microbiol 9: 1813) matches no record: Front Microbiol 9:1813 (doi:10.3389/fmicb.2018.01813) is an unrelated paper. Closest title: Berrios DC, et al. (2021) NASA GeneLab: interfaces for the exploration of space omics data. Nucleic Acids Res 49(D1): D1515–D1522,**

doi:10.1093/nar/gkaa887; its abstract does not support the Arabidopsis study count in the citing sentence.]

[3] **[TODO: unresolved citation. The original entry (Ray S, et al. (2022) NASA Open Science Data Repository: a comprehensive open data resource for biological studies in space. Life 12(11): 1859) matches no record: Life 12(11):1859 (doi:10.3390/life12111859) is an unrelated paper. Closest subject: Gebre SG, et al. NASA open science data repository: open science for life in space. Nucleic Acids Res 53(D1): D1697–D1710, doi:10.1093/nar/gkae1116; its abstract does not support the Arabidopsis study count in the citing sentence.]**

[4] Reddy ASN, Marquez Y, Kalyna M, Barta A (2013) Complexity of the alternative splicing landscape in plants. *Plant Cell* 25(10): 3657–3683. doi:10.1105/tpc.113.117523

[5] Marquez Y, et al. (2012) Transcriptome survey reveals increased complexity of the alternative splicing landscape in *Arabidopsis*. *Genome Res* 22(6): 1184–1195. doi:10.1101/gr.134106.111

[6] Staiger D, Brown JWS (2013) Alternative splicing at the intersection of biological timing, development, and stress responses. *Plant Cell* 25(10): 3640–3656. doi:10.1105/tpc.113.113803

[7] Filichkin S, Priest HD, Megraw M, Mockler TC (2015) Alternative splicing in plants: directing traffic at the crossroads of adaptation and environmental stress. *Curr Opin Plant Biol* 24: 125–135. doi:10.1016/j.pbi.2015.02.008

[8] Kalyna M, et al. (2012) Alternative splicing and nonsense-mediated decay modulate expression of important regulatory genes in *Arabidopsis*. *Nucleic Acids Res* 40(6): 2454–2469. doi:10.1093/nar/gkr932

[9] Drechsel G, et al. (2013) Nonsense-mediated decay of alternative precursor mRNA splicing variants is a major determinant of the *Arabidopsis* steady state transcriptome. *Plant Cell* 25(10): 3726–3742. doi:10.1105/tpc.113.115485

[10] Syed NH, et al. (2012) Alternative splicing in plants - coming of age. *Trends Plant Sci* 17(10): 616–623. doi:10.1016/j.tplants.2012.06.001

[11] Reddy ASN, Shad Ali G (2011) Plant serine/arginine-rich proteins: roles in precursor messenger RNA splicing, plant development, and stress responses. *Wiley Interdiscip Rev RNA* 2(6): 875–889. doi:10.1002/wrna.98

[12] NASA GeneLab. GeneLab RNA-seq Consensus Pipeline (GL-DPPD-7101-E). `https://github.com/nasa/GeneLab_Data_Processing`

[13] Wang Y, Xie Z, Kutschera E, Adams JI, Kadash-Edmondson KE, Xing Y (2024) rMATS-turbo: an efficient and flexible computational tool for alternative splicing analysis of large-scale RNA-seq data. *Nat Protoc* 19(4): 1083–1104. doi:10.1038/s41596-023-00944-2

[14] Rogers MF, et al. (2012) SpliceGrapher: detecting patterns of alternative splicing from RNA-Seq data in the context of gene models and EST data. *Genome Biol* 13(1): R4. doi:10.1186/gb-2012-13-1-r4

[15] **[TODO: unresolved citation. The original entry (Solheim BG, et al. (2021) Spaceflight-induced changes in expression of circadian clock genes in Arabidopsis. Life 11(8): 821) matches no record: Life 11(8):821 (doi:10.3390/life11080821) is an unrelated paper.]**